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United States Patent	12401157
Kind Code	B2
Date of Patent	August 26, 2025
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Reliable electrical connector with latch

Abstract

A card edge connector with latches that improve the connector's reliability. The connector comprises a housing having a slot for receiving a card and latches on opposite ends of the slot for locking the card in the slot. Each latch includes a body and a projection extending in a transverse direction from a side surface of the body. The latch is pivotably connected to the housing by the projection. The latch may rotate about the projection between a locked position and an unlocked position. The body includes a lower body and an upper body wider than the lower body in the transverse direction. The projection is disposed on the lower body. The latch further includes a protrusion extending between the projection and the upper body. Such a configuration restricts the latches from moving in the housing and therefore improves the reliability of the connector.

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Appl. No.: **17/973913**

Filed: **October 26, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230126150 A1	Apr. 27, 2023

Foreign Application Priority Data

CN	202122591708.7	Oct. 27, 2021

Publication Classification

Int. Cl.: **H01R13/629** (20060101); **H01R12/70** (20110101); **H01R12/72** (20110101)

U.S. Cl.:

CPC **H01R13/62988** (20130101); **H01R12/7005** (20130101); **H01R12/721** (20130101);

Field of Classification Search

CPC: H01R (12/7005); H01R (12/721); H01R (12/737); H01R (13/62988); H01R (13/635); H01R (13/6275)

USPC: 439/327

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Background/Summary

RELATED APPLICATIONS

(1) This application claims priority to and the benefit of Chinese Patent Application Serial No. 202122591708.7, filed on Oct. 27, 2021, entitled “ELECTRICAL CONNECTOR.” The contents of this applications are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(2) The embodiments of the present disclosure relate to an electrical connector.

BACKGROUND

(3) Electrical connectors are used in many electronic systems. It is generally easier and more cost effective to manufacture a system as several printed circuit boards (PCB) which may be joined together with electrical connectors than to manufacture the system as a single assembly. A known arrangement for interconnecting several PCBs is to have one PCB serve as a backplane. Other PCBs, which may be referred to as “daughter boards” or “daughter cards”, are connected through the backplane.

SUMMARY

(4) Aspects of the present disclosure relate to electrical connectors with improved reliability.

(5) Some embodiments relate to a latch for a card edge connector. The card edge connector may include a housing having a slot elongating in a longitudinal direction. The latch may include a body comprising an upper portion and a lower portion narrower than the upper portion in a transverse direction perpendicular to the longitudinal direction; a projection extending from the lower portion of the body in the transverse direction; and a protrusion extending from lower portion between the projection and the upper portion of the body.

(6) In some embodiments, the lower portion may have a first length in the longitudinal direction; and the protrusion may have a second length in the longitudinal direction, shorter than the first length.

(7) In some embodiments, the protrusion may comprise a bottom portion adjacent the projection and a top portion adjacent the upper portion of the body; and the top portion may extend beyond the bottom portion in the transverse direction.

(8) In some embodiments, the protrusion may comprise a bottom portion joining the projection and a top portion joining the upper portion of the body; and the top portion may extend beyond the bottom portion in the transverse direction.

(9) In some embodiments, the protrusion may comprise an intermediate portion joining the bottom portion and the top portion; and the intermediate portion may be narrower than the top portion and the bottom portion in the longitudinal direction.

(10) In some embodiments, the top portion may be flush with the upper portion of the body.

(11) In some embodiments, a lower part of the projection may diminish in size in the transverse direction along a vertical direction perpendicular to the transverse direction and the longitudinal direction.

(12) In some embodiments, the body may comprise a first side surface facing the slot of the connector and a second side surface opposite the first side surface and facing away from the slot of the connector; and the protrusion may be disposed closer to the first side surface than the second side surface.

(13) In some embodiments, the protrusion may comprise a first end surface facing the slot of the connector and a second end surface opposite the first end surface and facing away from the slot of the connector; and the second end surface may curve toward the slot.

(14) Some embodiments relate to an electrical connector. The electrical connector may include a housing comprising a slot elongated in a longitudinal direction; and a latch comprising a body and a projection extending in a transverse direction perpendicular to the longitudinal direction from a side surface of the body, the latch body pivotably connected to the housing by the projection such that the latch body is pivotable between a locked position and an unlocked position, and a

protrusion extending from the projection towards an upper portion of the body so as to be disposed in a gap between the side surface of the body and the housing.

(15) In some embodiments, the protrusion may include a bottom portion connected to the projection; and the projection may extend outwards from the body beyond the bottom portion in the transverse direction.

(16) In some embodiments, the protrusion may comprise a top portion connected to the upper portion of the body; and the top portion may extend outwards from the body beyond the bottom portion in the transverse direction to form an upper projecting platform between the top portion and the bottom portion.

(17) In some embodiments, the protrusion may comprise an intermediate portion between the top portion and the bottom portion; and the intermediate portion may have an outer surface smoothly connecting the top portion and the bottom portion.

(18) In some embodiments, the intermediate portion may be narrower than the bottom portion and the top portion in the longitudinal direction.

(19) In some embodiments, the housing may comprise a space configured to receive the latch; the space may comprise a side wall having a lower projecting platform; and the lower projecting platform may be disposed in a movement path of the upper projecting platform from the locked position to the unlocked position.

(20) In some embodiments, the protrusion may comprise a first end surface facing the slot and a second end surface facing away from the slot; and the second end surface is adjacent to and/or abuts against a portion on the housing when the body of the latch is in the unlocked position.

(21) Some embodiments relate to an electrical connector. The electrical connector may include a housing comprising a base extending in a longitudinal direction and a tower disposed at an end of the base and extending in a vertical direction perpendicular to the longitudinal direction; and a latch comprising a pair of projections pivotably disposed in the tower, a body extending in the vertical direction and rotatable about an axis connecting the pair of projections, and a pair of protrusions extending from respective projections of the pair and rotatably about the axis.

(22) In some embodiments, the pair of protrusions may extend from respective projections towards respective sides of an upper portion of the body.

(23) In some embodiments, the body of the latch may comprise an upper portion and a lower portion narrower than the upper portion in a transverse direction perpendicular to the longitudinal direction and the vertical direction; the pair of projections may extend from the lower portion of the body in the transverse direction; and the pair of protrusions may extend from respective projections of the pair to respective sides of the upper portion.

(24) In some embodiments, the body of the latch may comprise a through hole extending from the upper portion to the lower portion in the vertical direction.

(25) In some embodiments, the lower portion of the body may comprise a curved hook disposed in the base of the housing.

(26) In some embodiments, the latch and the housing may comprise matching portions configured to engage with each other when the latch is in a locked position.

(27) Some embodiments relate to an electrical connector. The electrical connector may comprise an insulating housing provided with a slot and a latch. The latch may include a latch body and a projection extending along a transverse direction from a side surface of the latch body. The latch body may be pivotably connected to the insulating housing by the projection between a locked position and an unlocked position. The latch may be used for locking a mated electronic element connected to the slot. The latch body may include a lower body and an upper body, the lower body may be narrower than the upper body in the transverse direction, and the projection may be provided on the lower body. The latch may further include a protrusion provided on the lower body, and the protrusion may extend from the projection towards the upper body and abut against the insulating housing in the transverse direction.

(28) In some embodiments, the protrusion may include a bottom portion connected to the projection, and the projection extends outwards from the latch body beyond the bottom portion in the transverse direction.
(29) In some embodiments, the protrusion may further include a top portion connected to the upper body, the top portion may extend outwards from the latch body beyond the bottom portion in the transverse direction to form a projecting platform between the top portion and the bottom portion.
(30) In some embodiments, an intermediate portion may connect between the top portion and the bottom portion, and the intermediate portion may have an outer surface smoothly connecting between the top portion and the bottom portion.
(31) In some embodiments, the intermediate portion may be narrower than the bottom portion and the top portion in a longitudinal direction perpendicular to the transverse direction.
(32) In some embodiments, the top portion may extend outwards along the transverse direction from the latch body to be flush with the upper body.
(33) In some embodiments, the insulating housing may be provided with a latch mounting slot, the latch may be pivotably mounted to the latch mounting slot, a side wall of the latch mounting slot may be provided with a lower projecting platform situated on a downward moving path of the upper projecting platform, there may be a gap between the lower projecting platform and the upper projecting platform, and the gap may be configured to be small enough to only allow the latch to be pivoted between the locked position and the unlocked position.
(34) In some embodiments, the side surface may have a first side facing the slot and a second side facing away from the slot, and the protrusion may be closer to the first side than the second side.
(35) In some embodiments, along a longitudinal direction perpendicular to the transverse direction, the protrusion may include a first end surface facing the slot and a second end surface facing away from the slot, the second end surface may abut against a limit portion on the insulating housing when the latch body is pivoted outward to the unlocked position.
(36) In some embodiments, the second end surface may be provided with a limit groove, the limit groove may abut against the limit portion when the latch body is situated at the unlocked position.
(37) In some embodiments, along the transverse direction, a lower part of the projection may have a diminishing size along a downward direction.
(38) In some embodiments, the insulating housing may include a base extending along the longitudinal direction and a tower protruding upward from an end of the base, and the latch body may be pivotably connected to the tower.
(39) In some embodiments, the projection may include a first projection and a second projection, the first projection and the second projection may be respectively provided on the two sides of the latch body opposite to each other along the transverse direction; wherein the protrusion may include a first protrusion portion and a second protrusion portion, the first protrusion portion may extend from the first projection towards the upper body, and the second protrusion portion may extend from the second projection towards the upper body.
(40) In some embodiments, the side surface may be provided with an engaging portion, the engaging portion may be situated at the upper body, the insulating housing may be provided with an engaging portion matching the engaging portion of the latch, and the engaging portion of the latch is engaged to or released from the engaging portions of the housing under an external force.
(41) In some embodiments, the engaging portion of the latch is a protrusion, and the engaging portion of the housing is an engaging groove.
(42) In some embodiments, the latch body may be provided with a through hole, the through hole may penetrate the latch body along a longitudinal direction perpendicular to the transverse direction.
(43) In some embodiments, the through hole may extend from the upper body to the lower body.
(44) In some embodiments, a lower end of the latch body may include a curved hook facing the slot, and the lower end of the latch body may abut against the insulating housing when the latch

body is situated at the unlocked position.
(45) These techniques may be used alone or in any suitable combination. The foregoing summary is provided by way of illustration and is not intended to be limiting.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) The accompanying drawings may not be drawn to scale. In the drawings, each identical or nearly identical component that is illustrated in various figures is represented by a like numeral. For purposes of clarity, not every component may be labeled in every drawing. In the drawings:
(2) FIG. **1** is a perspective view of an electrical connector according to an exemplary embodiment of the present disclosure, wherein the latch body is in the locked position;
(3) FIG. **2** is a perspective view of the electrical connector shown in FIG. **1**, wherein the latch body is in the unlocked position;
(4) FIG. **3** is a sectional perspective view of the electrical connector shown in FIG. **2**;
(5) FIG. **4** is a top perspective view of an insulating housing of the electrical connector of FIG. **1**;
(6) FIG. **5** is an enlarged view of a portion of the insulating housing circled in FIG. **4**;
(7) FIG. **6** is a side perspective view of the insulating housing shown in FIG. **4**;
(8) FIG. **7** is an enlarged view of a portion of the insulating housing circled in FIG. **6**;
(9) FIG. **8** is a side view of a latch of the electrical connector of FIG. **1**;
(10) FIG. **9** is a side view of the latch shown in FIG. **8**, showing a side of the latch facing a slot of the electrical connector of FIG. **1**;
(11) FIG. **10** is a perspective view of the latch shown in FIG. **8**, showing a surface of the latch facing away from a slot of the electrical connector of FIG. **1**;
(12) FIG. **11** is a perspective view of the latch shown in FIG. **8**, showing a surface of the latch facing a slot of the electrical connector of FIG. **1**; and
(13) FIG. **12** is a perspective view of the latch according to another exemplary embodiment of the present disclosure.
(14) The above accompanying drawings include the following reference signs:
(15) **100**-electrical connector; **110**-conductor; **200**-insulating housing; **201**-mating surface; **202**-mounting surface; **210**-receiving portion; **230**-limit portion; **240**-latch mounting slot; **241**, **242**-side wall; **243**-end wall; **244**-lower projecting platform; **251**-base; **252**-tower; **260**-engaging portion; **270**-projection receiver; **300**, **300′**-latch; **310**-latch body; **311**-lower body; **312**-upper body; **3120**-locking portion; **313**-lower end; **314**-upper end; **320**-projection; **321**-first projection; **322**-second projection; **330**-side surface; **331**-first side; **332**-second side; **340**-engaging portion; **350**-through hole; **360**-operation portion; **400**-protrusion; **411**-bottom portion; **412**-top portion; **413**-intermediate portion; **414**-upper projecting platform; **421**-first end surface; **422**-second end surface; **430**-limit groove; **441**-first protrusion portion; and **442**-second protrusion portion.
DETAILED DESCRIPTION
(16) The Inventors have recognized and appreciated design techniques to enable more reliable connectors. The Inventors have recognized and appreciated that conventional connectors may have latches that may move with respective to connector housings during mounting and mating, and/or in use, which may cause unreliable connections between components meant to be connected through the connectors. For example, a card edge connector is an example of a connector used for interconnection of printed circuit boards of an electronic system. A double data rate synchronous dynamic random access memory (DDR SDRAM) is an example of a memory used in computers. The DDR SDRAM may be interconnected to a mainboard of a computer through a card edge connector. The card edge connector may be mounted to the mainboard, and conductors of the card edge connector may be connected to circuits on the mainboard. A memory card may be inserted

into the card edge connector. The card edge connector may include a latch configured to lock the memory card with the card edge connector. Any movement of the latch may lead to movement of the memory card and therefore unreliable connection between the memory card and the mainboard.
(17) The inventors have recognized and appreciated features that prevent undesirable relative movements between components of connectors. A connector may have a housing with a base extending in a longitudinal direction and towers disposed on opposite ends of the base. Latches may be pivotably disposed in the towers so that the latches may move between a locked position for locking a mating component, such as a memory card, with the connector and an unlocked position for releasing the mating component from the connector. Each latch may have a body extending in a vertical direction and projections extending from opposite sides of the body in a transverse direction so as to be pivotably disposed in matching receivers of a respective tower. The body of the latch may rotate between the locked position and the unlocked position about an axis connecting the projections. The body of the latch may have an upper portion and a lower portion that may be narrower than the upper portion in the transverse direction such that the projections may extend from the lower portion without requiring enlarging the tower in the transverse direction. Gaps, however, may exist between the latch and the housing and give space for undesired movements of the latch.
(18) The latches may include features configured to prevent such undesired movements. Each latch may include protrusions configured to fill the gaps at least partially while still allowing the latches to move between the locked and unlocked positions. Each protrusion may have a bottom portion joining a respective projection and a top portion joining the upper portion of the body. The top portion may extend beyond the bottom portion in the transverse direction so as to prevent the latch from undesired movements in the transverse direction yet allow the movement between the locked and unlocked positions. The protrusion may be disposed closer to an inside of the tower and have a surface facing away the inside of the tower yet curving towards the inside of the tower so as to prevent the latch from undesired movements in the longitudinal direction yet allow the movement between the locked and unlocked positions. Each protrusion may include an intermediate portion joining the bottom portion and the top portion. The intermediate portion may be disposed adjacent a projecting platform of the tower so as to prevent the latch from undesired movements in the vertical direction due to, for example, a force generated during inserting a card or mounting the connector to a board, yet allow the movement between the locked and unlocked positions.
(19) Such techniques may be integrated into an electrical connector **100**. As shown in FIGS. **1** to **3**, the electrical connector **100** may include an insulating housing **200** and a latch **300**. For clear and concise description, a vertical direction Z-Z, a longitudinal direction X-X and a transverse direction Y-Y may be defined. The vertical direction Z-Z, the longitudinal direction X-X and the transverse direction Y-Y may be perpendicular to one another. The vertical direction Z-Z may refer to a height direction of the electrical connector **100**. The longitudinal direction X-X may refer to a length direction of the electrical connector **100**. The transverse direction Y-Y may refer to a width direction of the electrical connector **100**.
(20) The insulating housing **200** may have a mating surface **201** and a mounting surface **202** which may be opposite to each other in the vertical direction Z-Z. The mating surface **201** may be provided with a slot **210**. The slot **210** may have any suitable structure. In some embodiments, the slot **210** may be recessed inwards in the vertical direction Z-Z from the mating surface **201**, to form a card slot. The slot **210** may be configured to receive a mating electronic element. The mating electronic element may include, but not limited to, an electronic card. The electronic card may comprise one or more of a display card, a memory card, a sound card and the like. If possible, the mated electrical element may further include a mated electrical connector. For example, when the electrical connector **100** is a receptacle electrical connector, the mated electrical connector may be a plug electrical connector. The principle of the present disclosure is described below with an example of the mating electronic element being an electronic card.

(21) In some embodiments, the insulating housing **200** may have a longitudinal strip shape. The insulating housing **200** may extend in the longitudinal direction X-X. The slot **210** may be a long and thin card slot extending in the longitudinal direction X-X. The edge of the electronic card may be inserted into the card slot. The mounting surface **202** may face an element like a printed circuit board. The electronic card may be engaged to the slot **210** of the mating surface **201**, and the mounting surface **202** may be connected to a printed circuit board configured as a backplane. In this way, the electronic card may be electrically connected to the printed circuit board through the electrical connector **100**, and circuits on the electronic card and circuits on the printed circuit board may be interconnected.
(22) In some embodiments, as shown in FIGS. **3** to **7**, the insulating housing **200** may include a base **251** and a tower **252**. The base **251** may extend in the longitudinal direction X-X. The tower **252** may extend in the vertical direction Z-Z to protrude upward from an end of the base **251**. Orientation terms may be used herein under the placement of the electrical connector **100** shown in FIGS. **1** to **2**, i.e., a side on which the mating surface **201** may be located may be an upper side and a side on which the opposite mounting surface **202** may be located may be a lower side. Optionally, the receiving portion **210** in form of a card slot may be extend in the insulating housing **200** in the longitudinal direction X-X, and may extend, in the longitudinal direction X-X, into the tower **252** from the base **251**. The tower **252** may improve a connection strength of the electrical card inserted into the receiving portion **210**. The insulating housing **200** may be molded of an insulating material, for example, plastic. The insulating housing **200** may be an integral member.
(23) The base **251** may be provided with a plurality of conductors **110**. The plurality of conductors **110** may be spaced from each other in the longitudinal direction X-X to ensure mutual electric insulation of adjacent conductors **110**. Front ends of the plurality of conductors **110** may extend to the receiving portion **210**. In this manner, when the electrical card is engaged with the receiving portion **210**, the front ends of the conductors **110** may be electrically coupled with the conductors (e.g., golden fingers) on the electrical card. Rear ends of the plurality of conductors **110** may extend beyond the back surface (mounting surface) of the base **251**. When the electrical connector **100** is installed on the printed circuit board (not shown), the plurality of conductors **110** may be electrically coupled with the circuits on the printed circuit board.
(24) The latch **300** may be pivotably connected to an end of the insulating housing **200**. The latch **300** may be pivotably connected to the tower **252**. The latch **300** may be used for locking the electronic card connected to the receiving portion **210**. The edge portion, with the notch provide thereon, of the electronic card may be inserted into the tower **252**, and a locking portion **3120** (as shown in FIG. **11**) at the top of the latch **300** may protrude into the notch when the latch **300** is locked to the tower **252** such that the edge of the notch may be stuck in the locking portion **3120**. In some embodiments, the insulating housing may include a tower that does not bulge significantly. However, it should be appreciated that the latch **300** may be installed to the insulating housing by means of suitable variants. The insulating housing **200** may include the tower **252**. The lower portion of the side edge of the electrical card may be inserted into the insulating housing **200** to make the firmer connection between the electronic card and the electrical connector **100**. The tower **252** may further provide a mounting space for the latch **300**. In the embodiments shown in FIGS. **1** to **2**, in the longitudinal direction X-X, a pair of latches **300** may be disposed at outer ends of a pair of towers **252** respectively.
(25) As shown in FIGS. **8** to **11**, the latch **300** may include a latch body **310**, a projection **320** and a protrusion **400**. The projection **320** may extend, in the transverse direction Y-Y, from a side surface **330** of the latch body **310**. Projections **320** may be disposed on each of the side surface **330** of the latch body **310** and on the other side surface opposed to the side surface **330**. The projection **320** protrudes from the side surface **330** of the latch body **310** in the transverse direction Y-Y. The latch body **310** may be pivotably connected, enabling latch body **310** to pivot between a locked position and an unlocked position, to the insulating housing **200** by the projection **320**. The projection **320**

may be configured as a pivot for connecting the latch body **310** and the insulating housing **200**. When the latch body **310** is located at the locked position, the electronic card may be locked by the latch body **310** in the receiving portion **210**. When the latch body **310** is located at the locked position, the electronic card may be unlocked by pivoting the latch body **310** to the unlocked position such that the electrical card can be taken out of the receiving portion **210**. It should be appreciated that the insulating housing **200** may be provided with a projection receiver **270**, as shown in FIGS. **5** and **7**, to match the projection **320**, so that the latch body **310** may be pivotably connected, between the locked position and the unlocked position, to the insulating housing **200**. The projection receiver **270** may be a shaft hole or groove which may be disposed in the insulating housing **200**. The projection **320** may extend into the projection receiver **270**. The projection receiver **270** may have any structure capable of matching the projection **320** to enable the latch body **310** to be pivoted relative to the insulating housing **200**.
(26) In some embodiments, an upper end **314** of the latch body **310** may be provided with an operation portion **360**, as shown in FIG. **11**. The operation portion **360** may include one or more of anti-slip stripes and grooves. The operation portion **360** may make it convenient for users to pivot the latch body **310** between the locked position and the unlocked position, especially pivoting to the unlocked position.
(27) The latch body **310** may include a lower body **311** and an upper body **312**. In the transverse direction Y-Y, the lower body **311** may be narrower than the upper body **312**, as shown in FIG. **9**. In this manner, in the transverse direction Y-Y, an installation space may be formed on both sides of the lower body **311** for the projection **320** and the projection receiver **270**. The projection **320** may be disposed on the lower body **311**. In the vertical direction Z-Z, the projection **320** may be spaced apart from the upper body **312** to avoid an interference between the upper body **312** and the insulating housing **200** in the process of pivoting of the latch body **310**. The latch body **310** may be pivotably connected into a latch mounting slot **240** in the tower **252**, as shown in FIGS. **5** and **7**. In the transverse direction Y-Y, the width of the upper body **312** matches with that of an upper portion of the latch mounting slot **240**. When the latch body **310** may be located at the locked position, the upper body **312** may be clamped by two side walls **241** and **242** of the latch mounting slot **240**. In this manner, the latch **300** locked to the insulating housing **200** may have good stability, and the vibration of the latch **300** in the insulating housing **200** may be reduced.
(28) The protrusion **400** may be disposed on the lower body **311**. The protrusion **400** may extend, from the projection **320**, towards the upper body **312**. In some embodiments, in the longitudinal direction X-X, the size of the protrusion **400** may be generally no bigger than that of the projection **320**. In the longitudinal direction X-X, the size of the protrusion **400** may be slightly bigger than that of the projection **320**. Either of the above cases may be selected as long as the protrusion **400** does not interfere with the insulating housing **200** when the latch body **310** is pivoted between the locked position and the unlocked position. In some embodiments, as for the latch body **310** in the locked position, in the direction towards the longitudinal end of the insulating housing **200**, the protrusion **400** may not extend beyond the projection **320**. This may be mainly to avoid the interference with the insulating housing **200**. An end wall **243** of the tower **252** may be provided with an opening. The latch mounting slot **240** may be enclosed and formed by the end wall **243** and the side walls **241** and **242**. The upper body **312** may be pivoted to the outside of the insulating housing **200** in the longitudinal direction through the opening until the latch body **310** reaches the unlocked position, and accordingly, the locking portion **3120** on the upper body **312** configured to be locked with the electronic card may exit from the notch of the electronic card to leave an operation space for pulling out the electronic card from the electrical connector **100**. The protrusion **400** may be configured to have no interference with the end wall **243** in the process of the latch body **310** pivoting from the locked position to the unlocked position. Therefore, the protrusion **400** may be disposed closer to a longitudinal inner side of the lower body **311**.
(29) The protrusion **400** may be adjacent and/or abut against the insulating housing **200** in the

transverse direction Y-Y, and may be adjacent and/or abut against the side walls of the latch mounting slot **240**. This may facilitate locking the latch **300** in the insulating housing **200**. The protrusion **400** may stabilize the position of the electronic card relative to the electrical connector **100** by stabilizing the position of the latch **300** in the insulating housing **200**. The latch **300** may be prevented from vibrating in the insulating housing **200** in use, so that the electronic card may be stably held on the electrical connector **100**. The protrusion **400** may be also connected to the projection **320**, which may reinforce the mechanical strength of the projection **320**. In this way, the latch **300** may be firmer and more durable.

(30) The lower body **311** of the latch **300** may be provided with the protrusion **400**, such that the protrusion **400** and the insulating housing **200** may be adjacent to and/or abut against each other when the latch **300** is locked to the insulating housing **200**. In this manner, the latch **300** may not shake and have high stability, such that the electronic card held in the receiving portion **210** may have a relatively high stability, and the electrical connector **100** may have stable mechanical and electrical performances. The protrusion **400** may reinforce the mechanical strength of the projection **320**, which may make the latch **300** firmer and more durable.

(31) In some embodiments, in the transverse direction Y-Y, both sides of the latch body **310** may be provided with the projections **320** to improve the pivoting stability. As shown in FIGS. **8** to **11**, the projections **320** may include a first projection **321** and a second projection **322**. The first projection **321** and the second projection **322** may be disposed on two side surfaces of the latch body **310** opposed to each other in the transverse direction Y-Y respectively. The side wall **241** of the insulating housing **200** may be provided with a first projection receiver matching the first projection **321**, and the side wall **242** of the insulating housing **200** may be provided with a second projection receiver matching the second projection **322**. As described above, the first projection receiver and the second projection receiver may be shaft holes or grooves. The protrusion **400** may include a first protrusion portion **441** and a second protrusion portion **442**. The first protrusion portion **441** may extend, from the first projection **321**, towards the upper body **312**. The second protrusion portion **442** may extend, from the second projection **322**, towards the upper body **312**. The first protrusion portion **441** and the second protrusion portion **442** may be disposed on two sides of the latch body **310** opposed to each other in the transverse direction Y-Y respectively. The structures of the first protrusion portion **441** and the second protrusion portion **442** may be mirror images. Alternatively, the first protrusion portion **441** and the second protrusion portion **442** may have different structures. The first protrusion portion **441** and the second protrusion portion **442** may be adjacent to and/or abut against the insulating housing **200**, when the latch **300** is locked to the insulating housing **200**. Providing both sides with the protrusions may bring benefits. When the latch **300** is locked to the insulating housing **200**, the first protrusion portion **441** and the second protrusion portion **442** may be adjacent to and/or abut against the side walls **241** and **242** respectively. Therefore, the lower body **311** may be clamped between the side walls **241** and **242** by the first protrusion portion **441** and the second protrusion portion **442** to obtain a better stability.

(32) Although both sides of the latch body **310** may be provided with the projections **320**, the protrusion **400** may be provided on a single side. Providing a single side of the latch body **310** with the protrusion **400** may not be so good in improving the stability as the embodiments where the two sides of the latch body **310** may be provided with the protrusions **400**, but still provide benefits over conventional connectors.

(33) To further describe the structure of the protrusion **400** in details, the protrusion **400** will now be divided into a plurality of parts for purposes of discussion. As shown in FIGS. **8-10**, the protrusion **400** may include a bottom portion **411**. The bottom portion **411** may be connected to the projection **320**. In the transverse direction Y-Y, the projection **320** may extend outwards from the latch body **310** to exceed the bottom portion **411**. In this manner, the projection **320** may protrude from structures therearound and be inserted into the projection receiver **270** in form of a shaft hole to enable the matching between the projection **320** and the projection receiver **270** to be firmer, so

that the electrical connector **100** can be firmer and more durable.
(34) Further, as shown in FIGS. **8** to **10**, the protrusion **400** may further include a top portion **412**. The top portion **412** may be connected to the upper body **312**. The protrusion **400** may be connected between the upper body **312** and the projection **320**, which may enable the latch body **310** to have stronger mechanical strength. In other embodiments which may be not illustrated, the protrusion **400** may further be disposed with a spacing from the upper body **312**. In the transverse direction Y-Y, the top portion **412** may extend outwards from the latch body **310** to exceed the bottom portion **411**. Therefore, a projecting platform **414** may be formed between the top portion **412** and the bottom portion **411**, as shown in FIG. **9**. The top portion **412** may be adjacent to and/or abut against the insulating housing **200** in the transverse direction Y-Y. For example, the top portion **412** may be adjacent to and/or abut against the side wall of the latch mounting slot **240**. Although the top portion **412** exceeds the bottom portion **411** in the transverse direction Y-Y, the bottom portion **411** may be adjacent to and/or abut against the insulating housing **200** in the transverse direction Y-Y by reasonable design of the side wall of the latch mounting slot **240**, which therefore may bring a better stability effect.
(35) Also referring to FIG. **7**, the side wall of the latch mounting slot **240** may be provided with a projecting platform **244**. The projecting platform **244** may be situated on a downward moving path of the projecting platform **414**. When the latch **300** bears a downward pressing force and the projecting platform **414** moves downward to the position where the projecting platform **244** may be located, the projecting platform **244** may prevent the projecting platform **414** from continuing to move downward, so that the force applied to the projection **320** may be shared. When the latch **300** may be installed in place on the insulating housing **200**, there may be a gap between the projecting platform **244** and the projecting platform **414**. The gap may be configured to be small enough to only allow the latch **300** to pivot between the locked position and the unlocked position.
(36) When a downward pressing force is applied to the latch **300**, for example, when the latch **300** is installed downward, the protrusion **400** may be configured to support the projection **320**, so as to prevent any damage to the projection **320** by, for example, excessive force.
(37) In some embodiments, in the transverse direction Y-Y, the top portion **412** may extend outwards from the latch body **310** to be flush with the upper body **312**. In some embodiments, in the transverse direction Y-Y, an outer surface of the top portion **412** may be flush with an outer surface of the upper body **312**, as shown in FIG. **9**. In this manner, the latch **300** may have a relatively small size and a compact structure, so as to achieve miniaturization of the electrical connector **100**.
(38) As shown in FIGS. **8** to **10**, the protrusion **400** may include an intermediate portion **413**. The intermediate portion **413** may be between the top portion **412** and the bottom portion **411**. The intermediate portion **413** may have an outer surface smoothly connected between the top portion **412** and the bottom portion **411**. The smooth projecting platform **414** may be formed on the intermediate portion **413**. The projecting platform **244** may have a structure mated to the projecting platform **414**. The intermediate portion **413** may make the outer surface of the protrusion **400** smoother, so that a friction resistance of pivoting may be smaller, and user experience may be better. This latch **300** may be easy to manufacture.
(39) Alternatively, the top portion **412** and the bottom portion **411** may not be connected by a smooth curved surface, and the projecting platform **414** may be a step having a corner. Correspondingly, the projecting platform **244** may also be a step having a corner.
(40) As shown in FIG. **8**, in the longitudinal direction X-X, the intermediate portion **413** may be narrower than the bottom portion **411** and the top portion **412**. In this manner, a groove may be formed by the intermediate portion **413**, the bottom portion **411** and the top portion **412**. The groove may function as a limit groove **430** (described below).
(41) In some embodiments, as shown in FIG. **8**, the side surface **330** of the latch body **310** may have a first side **331** and a second side **332**. The first side **331** may face the receiving portion **210**,

with reference to FIGS. **1** to **3**. The second side **332** may face away from the receiving portion **210**. The protrusion **400** may be closer to the first side **331**, relative to the second side **332**. The protrusion **400** may be closer to the receiving portion **210**. In this manner, even though the latch body **310** may be pivoted to the unlocked position (as shown in FIG. **2**), the protrusion **400** may not have the interference with the tower **252**, and the protrusion **400** can nearly be in the tower **252** all the time. Therefore, the latch **300** may also have a relatively good stability during pivoting.
(42) In some embodiments, as shown in FIGS. **10** to **11**, in the longitudinal direction X-X, the protrusion **400** may include a first end surface **421** and a second end surface **422**. The first end surface **421** may face the receiving portion **210**. The second end surface **422** may face away from the slot **210**. The insulating housing **200** may be provided with a limit portion **230**. The second end surface **422** may be adjacent to and/or abut against the limit portion **230** on the insulating housing **200** when the latch body **310** may be pivoted outwards to the unlocked position. The limit portion **230** may include a stopper, a shutter, and the like. In this way, the latch body **310** may be prevented from damage due to, for example, excessive pivoting.
(43) The second end surface **422** may be provided with the limit groove **430**. The limit groove **430** may be adjacent to and/or abut against the limit portion **230** when the latch body **310** is located at the unlocked position. The limit groove **430** may be formed by having the intermediate portion **413** narrower than the bottom portion **411** and the top portion **412**.
(44) In some embodiments, as shown in FIG. **9**, in the transverse direction Y-Y, the lower part of the projection **320** may have a diminishing size in a downward direction. It may enable the projection **320** to have a guiding effect, facilitating the projection **320** to be inserted into the projection receiver **270** of the insulating housing **200** in form of the shaft hole.
(45) In some embodiments, as shown in FIG. **12**, the side surface **330** of the latch body **310** may be provided with engaging portions **340**, and the engaging portions **340** may be located at the upper body **312** of the latch body **310**. In addition to the engaging portions **340**, a latch **300′** in the embodiment shown in FIG. **12** may be generally the same as the latch **300** in the embodiment shown in FIGS. **8** to **11**. Therefore, the same reference numbers may be used for the same or similar parts. In some embodiments, the insulating housing **200** may be provided with engaging portions matching the engaging portions **340**, for example, engaging portions **260** shown in FIG. **7**. The engaging portions **260** may be located at positions, corresponding to the engaging portions **340**, on the side walls **241** and **242**. The engaging portions **340** may be engaged to the engaging portions **260** or released from the engaging portions **260** by, for example, an external force. The engaging portions **340** may be engaged to the engaging portions **260** when the latch body **310** is located at the locked position under reasonable configuration, such that the latch body **310** may be fixed at the locked position.
(46) Further, as shown in FIGS. **5** and **12**, the engaging portions **340** may be protrusions, and the engaging portions **260** may be engaging grooves. In this manner, the engaging portions **340** and the engaging portions **260** have simple structures to facilitate manufacturing.
(47) In some embodiments, as shown FIGS. **9** to **11**, the latch body **310** may be provided with a through hole **350**. The through hole **350** may penetrate the latch body **310** in the longitudinal direction X-X. The number of the through hole **350** may be one or more. The through hole **350** may facilitate heat dissipation of the electronic card, so that the electronic card may be prevented from being damaged due to excessive heat. Further, as shown in FIGS. **9** to **11**, the through hole **350** may extend from the upper body **312** to the lower body **311**. In this manner, the through hole **350** has a bigger size for a better heat dissipation effect.
(48) In some embodiments, as shown in FIGS. **10** to **11**, a lower end **313** of the latch body **310** may include a curved hook facing the receiving portion **210**. The lower end **313** of the latch body **310** may be adjacent to and/or abut against the insulating housing **200** when the latch body **310** is located at the unlocked position. Therefore, the lower end **313** of the latch body **310** may be used for position limiting.

(49) Although the present disclosure has been described through the above embodiments, it should be appreciated that the above embodiments are only for the purpose of illustration and description, and are not intended to limit the present disclosure to the scope of the described embodiments. In addition, it may be understood by a person skilled in the art that the present disclosure is not limited to the above embodiments, a variety of variations and modifications may be made according to the teaching of the present disclosure, and these variations and modifications all fall within the scope of protection of the present disclosure. The scope of protection of the present disclosure is defined by the appended claims and its equivalent scope.
(50) Various changes may be made to the illustrative structures shown and described herein. For example, although the latch described above may be used in a card edge connector, the latch may be used in any suitable electrical connector, such as backplane connectors, daughter card connectors, stacking connectors, Mezzanine connectors, I/O connectors, chip sockets, Gen Z connectors, etc.
(51) Moreover, although many creative aspects have been described above with reference to the right angle connector, it should be understood that the aspects of the present disclosure are not limited to these. Any one of the creative features, whether alone or combined with one or more other creative features, can also be used for other types of electrical connectors, such as coplanar connectors.
(52) In the description of the present disclosure, it is to be understood that orientation or positional relationships indicated by orientation words “front”, “rear”, “upper”, “lower”, “left”, “right”, “transverse direction”, “vertical direction”, “perpendicular”, “horizontal”, “top”, “bottom” and the like usually are shown based on the accompanying drawings, only for the purposes of the ease in describing the present disclosure and simplification of its descriptions. Unless stated to the contrary, these orientation words do not indicate or imply that the specified apparatus or element has to be specifically located, and structured and operated in a specific direction, and therefore, should not be understood as limitations to the present disclosure. The orientation words “inside” and “outside” refer to the inside and outside relative to the contour of each component itself.
(53) For facilitating description, the spatial relative terms such as “on”, “above”, “on an upper surface of” and “upper” may be used here to describe a spatial position relationship between one or more components or features and other components or features shown in the accompanying drawings. It should be understood that the spatial relative terms not only include the orientations of the components shown in the accompanying drawings, but also include different orientations in use or operation. For example, if the component in the accompanying drawings is turned upside down completely, the component “above other components or features” or “on other components or features” will include the case where the component is “below other components or features” or “under other components or features”. Thus, the exemplary term “above” can encompass both the orientations of “above” and “below”. In addition, these components or features may be otherwise oriented (for example rotated by 90 degrees or other angles) and the present disclosure is intended to include all these cases.
(54) It should be noted that the terms used herein are only for describing specific embodiments, and are not intended to limit the exemplary embodiments according to the present application. As used herein, an expression of a singular form includes an expression of a plural form unless otherwise indicated. In addition, the use of “including”, “comprising”, “having”, “containing”, or “involving”, and variations thereof herein, is meant to encompass the items listed thereafter (or equivalents thereof) and/or as additional items.
(55) It should be noted that the terms “first”, “second” and the like in the description and claims, as well as the above accompanying drawings, of the present disclosure are used to distinguish similar objects, but not necessarily used to describe a specific order or precedence order. It should be understood that ordinal numbers used in this way can be interchanged as appropriate, so that the

embodiments of the present disclosure described herein can be implemented in a sequence other than those illustrated or described herein.

Claims

1. A latch for a card edge connector, the card edge connector comprising a housing having a slot elongating in a longitudinal direction, the latch comprising: a body comprising an upper portion and a lower portion narrower than the upper portion in a transverse direction perpendicular to the longitudinal direction; a projection extending from the lower portion of the body in the transverse direction; and a protrusion extending from the lower portion between the projection and the upper portion of the body, the protrusion comprising: a bottom portion, a top portion, and an intermediate portion narrower in the longitudinal direction than the top portion and the bottom portion.
2. The latch of claim 1, wherein: the lower portion has a first length in the longitudinal direction; and the protrusion has a second length in the longitudinal direction, shorter than the first length.
3. The latch of claim 2, wherein: the bottom portion is adjacent the projection; the top portion is adjacent the upper portion of the body; and the top portion extends beyond the bottom portion in the transverse direction.
4. The latch of claim 1, wherein: the bottom portion is joined to the projection; the top portion is joined to the upper portion of the body; and the top portion extends beyond the bottom portion in the transverse direction.
5. The latch of claim 4, wherein: the top portion is flush with the upper portion of the body.
6. The latch of claim 1, wherein: a lower part of the projection diminishes in size in the transverse direction along a vertical direction perpendicular to the transverse direction and the longitudinal direction.
7. The latch of claim 1, wherein: the body comprises a first side surface facing the slot of the connector and a second side surface opposite the first side surface and facing away from the slot of the connector; and the protrusion is disposed closer to the first side surface than the second side surface.
8. The latch of claim 7, wherein: the protrusion comprises a first end surface facing the slot of the connector and a second end surface opposite the first end surface and facing away from the slot of the connector; and the second end surface curves toward the slot.
9. An electrical connector, comprising: a housing comprising a slot elongated in a longitudinal direction; and a latch comprising: a body comprising a first side surface facing the slot of the housing, a second side surface opposite the first side surface and facing away from the slot of the housing, and a third side surface between the first side surface and the second side surface, a projection extending in a transverse direction perpendicular to the longitudinal direction from the third side surface of the body, the body of the latch pivotably connected to the housing by the projection such that the body of the latch is pivotable between a locked position and an unlocked position, and a protrusion extending from the projection towards an upper portion of the body so as to be disposed in a gap between the third side surface of the body and the housing, the protrusion comprising a first end surface facing the slot of the housing and a second end surface opposite the first end surface and facing away from the slot of the housing, the second end surface curving toward the slot.
10. The latch electrical connector of claim 9, wherein: the protrusion is disposed closer to the first side surface of the body of the latch than the second side surface of the body of the latch.
11. The electrical connector of claim 9, wherein: the protrusion includes a bottom portion connected to the projection; and the projection extends outwards from the body beyond the bottom portion in the transverse direction.
12. The electrical connector of claim 11, wherein: the protrusion comprises a top portion connected to the upper portion of the body; and the top portion extends outwards from the body beyond the

bottom portion in the transverse direction to form an upper projecting platform between the top portion and the bottom portion.

13. The electrical connector of claim 12, wherein: the protrusion comprises an intermediate portion between the top portion and the bottom portion; and the intermediate portion has an outer surface smoothly connecting the top portion and the bottom portion.

14. The electrical connector of claim 13, wherein: the intermediate portion is narrower than the bottom portion and the top portion in the longitudinal direction.

15. The electrical connector of claim 12, wherein: the housing comprises a space configured to receive the latch; the space comprises a side wall having a lower projecting platform; and the lower projecting platform is disposed in a movement path of the upper projecting platform from the locked position to the unlocked position.

16. The electrical connector of claim 9, wherein: the protrusion comprises a first end surface facing the slot and a second end surface facing away from the slot; and the second end surface is adjacent to and/or abuts against a portion on the housing when the body of the latch is in the unlocked position.

17. An electrical connector comprising: a housing comprising a base extending in a longitudinal direction and a tower disposed at an end of the base and extending in a vertical direction perpendicular to the longitudinal direction; and a latch comprising a pair of projections pivotably disposed in the tower, a body extending in the vertical direction and rotatable about an axis connecting the pair of projections, and a pair of protrusions extending from respective projections of the pair and rotatably about the axis, wherein each of the pair of protrusions comprise a bottom portion, a top portion, and an intermediate portion narrower in the longitudinal direction than the top portion and the bottom portion.

18. The electrical connector of claim 17, wherein: the pair of protrusions extend from respective projections towards respective sides of an upper portion of the body.

19. The electrical connector of claim 17, wherein: the body of the latch comprises an upper portion and a lower portion narrower than the upper portion in a transverse direction perpendicular to the longitudinal direction and the vertical direction; the pair of projections extend from the lower portion of the body in the transverse direction; and the pair of protrusions extend from respective projections of the pair to respective sides of the upper portion.

20. The electrical connector of claim 19, wherein: the body of the latch comprises a through hole extending from the upper portion to the lower portion in the vertical direction.

21. The electrical connector of claim 19, wherein: the lower portion of the body comprises a curved hook disposed in the base of the housing.

22. The electrical connector of claim 17, wherein: the latch and the housing comprise matching portions configured to engage with each other when the latch is in a locked position.